

Tutorial Problem Set III

MATA30 – TUT0003

OLLE RENÉRIUS REHNQUIST

January 23, 2026

In the first hour of this tutorial, you will work on problems related to inverse functions, exponential and logarithmic functions and hyperbolic functions. In the second hour, you will complete Quiz 1.

1 Basic problems

Problem 1. Trigonometric and hyperbolic functions.

- (a) Compute $\sin^{-1}(\sin(\pi))$. Be careful!
- (b) Recall that we have the following trigonometric identities:

$$\cos^2(x) + \sin^2(x) = 1 \quad \text{and} \quad \cos^2(x) - \sin^2(x) = \cos(2x).$$

Prove the corresponding *hyperbolic* identities:

$$\cosh^2(x) + \sinh^2(x) = \cosh(2x) \quad \text{and} \quad \cosh^2(x) - \sinh^2(x) = 1.$$

- (c) Prove that for all nonzero x :

$$\csc(\tan^{-1}(x)) = \frac{\sqrt{1+x^2}}{x}.$$

Problem 2. Inverse functions.

- (a) Show that the function $f(x) = \sinh(x) = \frac{e^x - e^{-x}}{2}$ is one-to-one and that $\text{Range}(f) = \mathbb{R}$. Find f^{-1} .
- (b) Show that the function $f(x) = e^{x^2-2x} + 1$ is not one-to-one. Can you find a restricted domain where f is one-to-one? Compute the inverse of f there and give its domain.

2 Challenging problems

Problem 3. Find all $x \in \mathbb{R}$ such that

$$x(2x^2 - 1)(8x^4 - 8x^2 + 1) = \frac{1}{8}.$$

Hint: There exists an angle θ such that $x = \cos(\theta)$.

Problem 4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)f(y)) = f(x)y$$

for all $x, y \in \mathbb{R}$. *Hint:* Prove first that any such function is one-to-one.